SPECIMEN INTEGRITY

Introduction

Within the clinical laboratory setting, certified phlebotomists are responsible for specimen collection, preparation, and transportation. The following sections will discuss the key important steps to preparing a specimen for testing or transport, maintaining specimen integrity, and adhering to the ***chain of custody*** for forensic specimens.

chain of custody. A special protocol that is followed when forensic specimens are collected.

Preparing Specimens for Testing or Transport

Specimens must be handled properly after being collected from the patient. Improperly handled specimens can impact the patient's results. Therefore, transporting and testing each specimen as quickly as possible will provide results that accurately represent the patient's condition at the time of collection. Here are some guidelines to keep in mind while preparing specimens for testing and transport.

- Refrigeration and freezing: The storage temperature of a specimen should follow the manufacturer's guidelines.
- Centrifuging: Serum should be separated from red blood cells as soon as possible after the specimen has clotted to prevent changes caused by the ***hemolysis*** of the blood cells.
- Some specimens must be refrigerated or frozen to prevent chemical constituents from changing.
- The manufacturer's insert or reference laboratory specimen requirements should be checked to make sure the specimen is handled and processed correctly.

hemolysis. Damage to the RBCs results in the leakage of hemoglobin into the fluid part of the specimen.

Major laboratories typically feature a specific section known as central/specimen processing. This area is dedicated to the reception and preparation of specimens for analysis. Upon arrival, specimens undergo a process of identification, logging, and categorization based on the department they are destined for and the processing they require. Maintain the proper conditions for each specimen in the central/specimen processing area, such as refrigeration, shielding from light, or ensuring they remain warm until they are ready for testing.

Specimens that meet the necessary criteria are then analyzed and assessed for further processing. Laboratories are responsible for verifying that each specimen received has been correctly identified and collected in the appropriate container at the specified time and under any special collection conditions. Additionally, all specimens must be filled to a satisfactory level and transported under the right conditions without delay. Specimens not meeting these standards may be deemed unsuitable for testing and consequently rejected, necessitating a request for a new sample. The most frequent cause for the rejection of chemistry specimens is hemolysis, which is the breakdown of red blood cells, followed by an inadequate volume of blood in the sample. In hematology, the primary reason for specimen rejection is the clotting of the blood sample.

Centrifuging

After collection, a blood specimen may be required to be processed before testing. This may include separating the plasma or serum from the red blood cells in a ***centrifuge*** machine. The preparation of a specimen involves three stages: the pre-centrifugation stage (which occurs between the time of collection and before the specimen is placed in a centrifuge), the centrifugation stage (where the specimen is spun in a centrifuge), and the post-centrifugation stage (which takes place after centrifugation and before the serum or plasma is extracted). Specimens arriving from external sites may have already undergone centrifugation, allowing them to move directly to the post-centrifugation steps once all necessary specimen criteria are satisfied. Throughout all three stages of centrifugation, it is imperative to maintain the tubes in a vertical orientation and ensure the stoppers remain securely in place. Removing the stopper can cause a loss of carbon dioxide from the specimen and an increase in pH levels, which may result in test results that are not accurate.

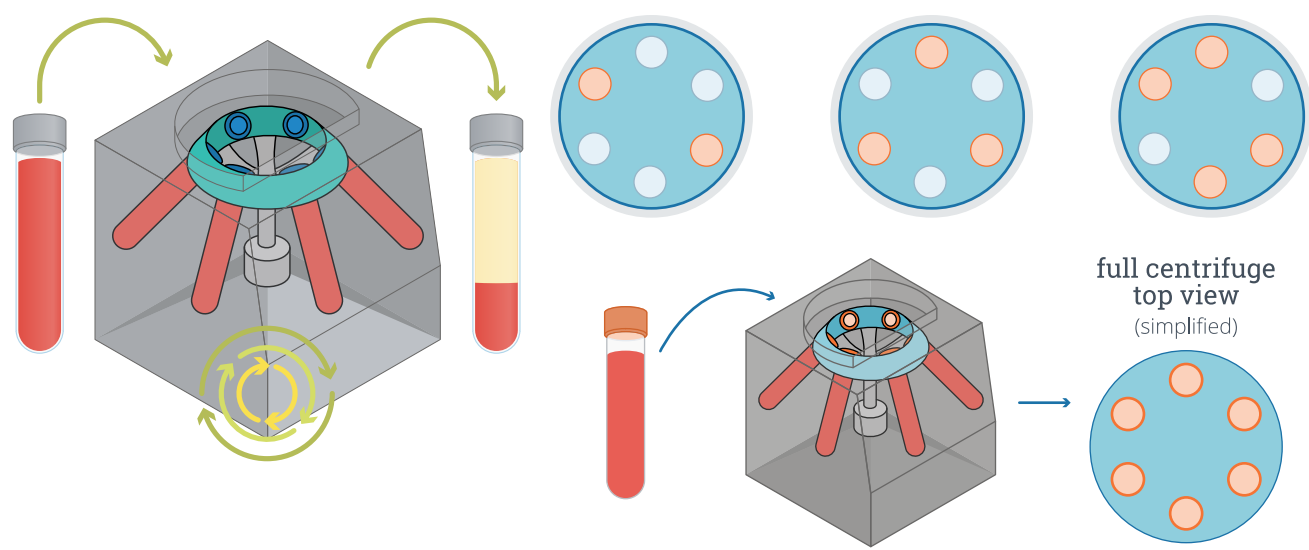
Do Not Remove the Stopper

Removing the stopper during centrifugation will cause the loss of carbon dioxide and increase the pH of the specimen, leading to inaccurate test results.

Do Not Double Centrifuge a Specimen

Specimens should not be centrifuged more than once. Multiple centrifugations can lead to hemolysis and inaccurate test results. Once the serum or plasma has been removed, the volume ratio changes, leading to inaccurate test results.

centrifuge. A machine that spins the blood tubes at the high number of revolutions per minute.



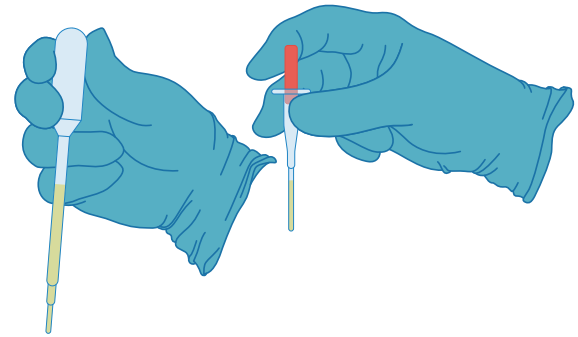
Centrifuging can be started when it has been determined that the clotting of the specimen is complete. If a specimen is not completely clotted, it could lead to residual fibrin that can alter test results or result in problems with the instrumentation and downtime of equipment, leading to delays in testing. Complete clotting times take approximately 30 to 60 min at room temperature. Patients taking anticoagulant medications, specimens with high white blood counts, or chilled specimens may take a bit longer to clot. A completely clotted specimen can be determined by gently inverting the tube to check if a solid clot has formed. Centrifugation should not be done more than once on a specimen. The repetition of centrifuging can lead to hemolysis and alter the test results.

The operation of the centrifuge is conducted while keeping the area clear of clutter around the centrifuge. For a centrifuge to operate effectively, it must be put in an area that allows adequate ventilation. Before operating the centrifuge, observing the specimen tubes for any cracks or imperfections that may result in breakage and exposure to hazardous body fluids is important. The specimen must also be verified to be completely clotted.

A critical component when processing a specimen is to balance the specimen when placing it into the centrifuge. This means placing equivalent-sized tubes with equal volumes on the opposite side of the centrifuge. Every tube must have a balance tube. If preparing a single-tube specimen, balance the specimen tube with a tube of the same size filled with water of the same volume as the specimen tube. When operating a centrifuge, any excessive shaking, noisier than usual, or excessive vibrating may indicate that the centrifuge is not balanced correctly and should be turned off immediately.

Prior to operating the centrifuge, it is essential to check that the timer and rotational speed are adjusted accurately. An excessively long centrifugation time or an overly high speed can lead to hemolysis of the specimen. Conversely, a duration that is too brief or a speed that is too low may prevent proper separation of the specimen. In case of a centrifuge malfunction during operation, promptly switch off the machine, disconnect it from the power source, and alert a supervisor. The centrifuge should not be used until it has been thoroughly inspected and confirmed to function correctly or repaired.

Aliquoting



After the centrifugation process is finished, the serum or plasma must be fully separated from the blood cells. The specimen is then divided into ***aliquot*** tubes. An aliquot refers to a specific portion of the specimen designated for testing. Using aliquots is common practice when several tests are required from one specimen, and those tests must be conducted on various instruments or in separate areas of the laboratory. To prepare an aliquot tube, a portion of the serum or plasma is transferred from the original centrifuged specimen into clearly marked tubes that carry the same patient identification as the original specimen tube.

A certified phlebotomist must take extra care when transferring the serum or plasma to the designated aliquot tubes. Compare two patient identifiers from the primary tube to the patient identifiers on the aliquot tubes. Serum and plasma are indistinguishable when transferred to aliquot tubes. All aliquot tubes must be covered or capped when filled with serum or plasma. CLSI recognizes that the serum or plasma aliquots can stay at room temperature for no longer than 8 hr. If the tests are not completed in 8 hr, the serum or plasma should be refrigerated at 2° C to 8° C (35.6° F to 46.4° F). If the tests are not being conducted immediately or the serum or plasma must be stored for 48 hr, it should be frozen below –20° C (–4° F). Some tests require refrigeration or freezing for ***analyte*** stability. Frozen samples must be thawed at room temperature and inverted 10 to 20 times after thawing before testing.

aliquot. A portion of the specimen used for testing.

Caution When Freezing or Refrigerating Aliquot

Aliquots of serum or plasma should undergo freezing and thawing only once. Repeated freezing can degrade the analytes that must be measured. Additionally, using "frost-free" freezers should be avoided as they can cause specimens to partially thaw and refreeze, compromising the sample integrity.

analyte. Term that refers to the substance undergoing analysis.

Caution When Transferring Serum and Plasma

Always use a pipette when transferring the serum or plasma into the aliquot tubes, as directly pouring could heighten the risk of creating aerosols or causing splashes. Additionally, it is important not to combine serum or plasma from specimens treated with different anticoagulants into a single aliquot tube.

Refrigerating and Freezing Specimens

It is not abnormal for specimens to be kept for a period before batch analysis or before being sent out. Typically, specimens are refrigerated after analysis in case other tests are ordered later. Most chemistry tests can be performed on separated serum or plasma for up to a week. The specimens must have a cover and be stored in an upright position. Typically, blood specimens can be stored and refrigerated at 2° C to 8° C (35.6° F to 46.4° F) for approximately 72 hr without the cells breaking down. While refrigeration is preferred, some specimens may require freezing. Whole blood samples can be stored for long-term usage by freezing, but a freezing temperature of –20° C (–4° F) or lower must be maintained.

6.1 CHALLENGE

1. How you handle a specimen can depend on the type of test you are performing. You have the choice of two handling options. Drag the type of test to the type of handling that would be required for any specimens.
   1. Biohazard bag for photsensitive specimens
   2. Ice slurry

A. Ammonia
B. Vitamin A
C. Bilirubin
D. Folate
E. Lactic Acid

6.1 CHALLENGE KEY

1. Biohazard bag for photsensitive specimens: B, C, D
   Ice slurry: A, E